

Validating Fairness: A Causal Inference Pipeline for Discriminatory Bias Assessment

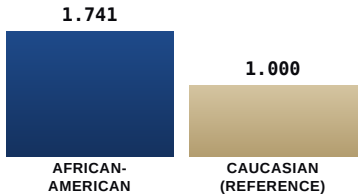
Correlation reveals **that** disparity exists. Causal discovery reveals **where** it comes from — and **how** to intervene. Demonstrated on ProPublica COMPAS (n=5,278).

STEP 01 — DETECT

Establish the Correlation Baseline

Compute correlation-based fairness metrics (Disparate Impact Ratio, selection-rate gap, rate ratios) to confirm a disparity is present.

SELECTION-RATE DISPARITY (COMPAS)



COMPAS CORRELATION METRICS

Selection-Rate DIR	1.7410
Recidivism Rate Ratio	1.3380
4/5 Rule (inverted)	0.575 FAILS

Correlation detects that bias exists — but offers no mechanistic explanation of its source.

STEP 02 — DISCOVER

Select a Causal Discovery Algorithm

Six algorithms across four families were benchmarked on synthetic data with planted ground-truth bias ($\beta = -0.15$).

ALGORITHM FAMILIES TESTED

CONSTRAINT-BASED PC · FCI

Detection < 50% at all (n, β); systematic mis-orientation.

SCORE-BASED (BIC) GES

Marginal: ~70% only at larger β and n .

PERMUTATION GRaSP

Detects planted bias at $n \geq 5,000$ (SHD=3), but unstable on unbiased data (SHD=11) and fails on real-world scores ("black-swan", Step 3).

FUNCTIONAL CAUSAL **PREFERRED**

ICA-LiNGAM · DirectLiNGAM
 $\geq 85\%$ detection at $n=1,000$; recovers oriented edges and β magnitudes.

WHY DIRECTLINGAM FOR REAL-WORLD ASSESSMENT

Exploits non-Gaussianity · guaranteed convergence · no ICA hyperparameters · robust to binary & heavy-tailed variables typical of high-stakes administrative data.

Algorithm choice is not methodologically neutral — it determines which forms of discriminatory bias remain visible.

STEP 03 — ISOLATE

Isolate Direct vs. Proxy Discrimination

Map the recovered DAG to separate two structurally distinct discrimination pathways — each demanding a different intervention.

► DIRECT PATHWAY

Race → Score

Discriminatory bias embedded *inside* the scoring rules themselves. **Intervention:** revise the algorithm's internal logic.

► PROXY PATHWAY

Race → Priors → Score

Structural inequities (e.g., racially disparate policing) inflate a "neutral" risk signal. **Intervention:** upstream policy reform — removing Race alone will not eliminate this path.

WATCH FOR: "BLACK SWAN" FAILURE MODES

BIC-ordering heuristics (e.g., GRaSP) can misclassify high- R^2 outcomes as exogenous — producing implausible arrows *into* immutable attributes like **Race** or **Sex**. Avoid purely BIC-driven permutation search when scores are highly collinear with their inputs.

STEP 04 — QUANTIFY

Quantify the Causal Effect

Apply Pearl's backdoor adjustment to estimate the Average Treatment Effect (ATE) — controlling for legitimate covariates {Age, Charge Degree, Juv Fel, Juv Misd, Priors}.

ATE CONVERGENCE UNDER CONTROLS

ATE (raw)		+1.6416
ATE (partial)		+1.1386
ATE (full)		+0.5255
DirectLiNGAM β		+0.5167
ICA-LiNGAM β		+1.2105

$\Delta = 0.0088$
CONVERGENT CROSS-VALIDATION

ATE (regression) and DirectLiNGAM (non-Gaussianity) — methods sharing *no* assumptions beyond linearity — agree within **0.0088** standard-deviation units, isolating discriminatory bias from legitimate risk signals.

A residual gap survives full criminal-history controls — consistent with direct discriminatory bias.

STEP 05 — VALIDATE

Validate with Bootstrapping

1,000 non-parametric bootstrap resamples (fixed seed) generate percentile-based 95% confidence intervals to test statistical stability.

95% CI · DIRECTLINGAM RACE → COMPAS SCORE



ALL THREE CIS EXCLUDE ZERO

DirectLiNGAM 95% CI	[+0.385, +0.637]
ATE (full) 95% CI	[+0.395, +0.663]
ICA-LiNGAM 95% CI	[+1.069, +1.345]
E-value (sensitivity)	≈ 1.7

Reject the null of zero bias.

Convergent point estimates, overlapping CIs excluding zero, and an E-value comparable to the strongest known criminogenic risk factors make a confounding explanation unlikely.

STEP 01 Detect

Does disparity exist?

STEP 02 Discover

Which algorithm can see it?

STEP 03 Isolate

Direct or via a proxy?

STEP 04 Quantify

How large, after controls?

STEP 05 Validate

Stable & not by chance?